



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

22MA201

ENGINEERING MATHEMATICS II

UNIT IV

VECTOR INTEGRAL CALCULUS

1. SURFACE INTEGRAL OF VECTOR POINT FUNCTIONS

1.1 INTRODUCTION

The line integral of a vector field F could be interpreted as the work done by the force field F on a particle moving along the path. The surface integral of a vector field F actually has a simpler explanation. If the vector field F represents the flow of a fluid, then the surface integral of F will represent the amount of fluid flowing through the surface (per unit time).

The amount of the fluid flowing through the surface per unit time is also called the **flux** of fluid through the surface. For this reason, we often call the surface integral of a vector field a **flux integral**.

If water is flowing perpendicular to the surface, a lot of water will flow through the surface and the flux will be large. On the other hand, if water is flowing parallel to the surface, water will not flow through the surface, and the flux will be zero. To calculate the total amount of water flowing through the surface, we want to add up the component of the vector F that is perpendicular to the surface.

1.1.1 SURFACE INTEGRAL

Consider a continuous function F and a surface S . Divide S into finite number of sub surfaces. Let the surface element surrounding any point P be δS which can be regarded as a vector; its magnitude being the area and its direction that of the outward normal to the element.

Consider the sum $\sum F \cdot \delta S$, where the summation extends over all the sub-surfaces. The limit of this sum as the number of sub surfaces tends to infinity and the area of each sub surface tends to zero, is called the **normal surface integral** of **F** over S and it is denoted by

$$\iint_S \vec{F} \cdot d\vec{S} \text{ or } \iint_S \vec{F} \cdot \hat{n} ds \quad \text{where } \hat{n} \text{ is a unit outward normal at P to}$$

S.

1.1.2 FLUX ACROSS A SURFACE

If **F** represents the velocity of a fluid particle, then the total outward flux of **F** across a

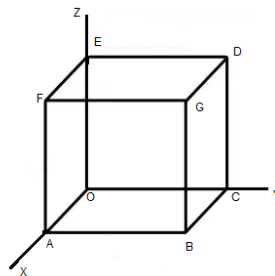
closed surface S is the surface integral $\iint_S \vec{F} \cdot d\vec{S}$

1.1.3 PROBLEMS

Example 1:

Find $\iint_S \vec{F} \cdot \hat{n} ds$ for $\vec{F} = 4xz \vec{i} - y^2 \vec{j} + yz \vec{k}$ and s is the surface of the cube bounded by the panes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

Solution:



Surface	Equation	$\hat{n}$	$\vec{F}$	$\vec{F} \cdot \hat{n}$	dS
S_1 OABC	$z = 0$	$-\vec{k}$	$-y^2 \vec{j}$	0	$dx dy$
S_2 EFGD	$z = 1$	$\vec{k}$	$4x \vec{i} - y^2 \vec{j} + y \vec{k}$	y	$dx dy$
S_3 OCDE	$x = 0$	$-\vec{i}$	$-y^2 \vec{j} + yz \vec{k}$	0	$dy dz$

S_4 ABGF	$x = 1$	$\vec{i}$	$4z \vec{i} - y^2 \vec{j} + yz \vec{k}$	$4z$	$dydz$
S_5 OEFA	$y = 0$	$-\vec{j}$	$4xz \vec{i}$	0	$dx dz$
S_6 BCDG	$y = 1$	$\vec{j}$	$4xz \vec{i} - \vec{j} + z \vec{k}$	-1	$dx dz$

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} \vec{F} \cdot \hat{n} \, ds + \iint_{S_2} \vec{F} \cdot \hat{n} \, ds + \iint_{S_3} \vec{F} \cdot \hat{n} \, ds + \iint_{S_4} \vec{F} \cdot \hat{n} \, ds + \iint_{S_5} \vec{F} \cdot \hat{n} \, ds + \iint_{S_6} \vec{F} \cdot \hat{n} \, ds$$

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} \, ds &= \int_0^1 \int_0^1 y \, dx \, dy + \int_0^1 \int_0^1 4z \, dy \, dz + \int_0^1 \int_0^1 -1 \, dx \, dz \\ &= \int_0^1 \left(\frac{y^2}{2} \right)_0^1 dx + 4 \int_0^1 \left(\frac{z^2}{2} \right)_0^1 dy - \int_0^1 (x)_0^1 dz \\ &= \frac{1}{2} (x)_0^1 + 2(y)_0^1 - (z)_0^1 \\ &= \frac{3}{2}. \end{aligned}$$

Example 2:

Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ if $\vec{F} = \vec{i}(x + y^2) - 2x\vec{j} + 2yz\vec{k}$ and S is the surface of the plane $2x + y + 2z = 6$

in the first octant.

Solution:

Given, $\vec{F} = (x + y^2)\vec{i} - 2x\vec{j} + 2yz\vec{k}$

$$\phi = 2x + y + 2z - 6$$

$$\nabla\phi = 2\vec{i} + \vec{j} + 2\vec{k}$$

$$|\nabla\phi| = \sqrt{4+1+4} = 3$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2\vec{i} + \vec{j} + 2\vec{k}}{3}$$

Let R_1 be the projection of S on the xy plane

$$\therefore \iint \vec{F} \cdot \hat{n} ds = \iint_{R_1} \vec{F} \cdot \hat{n} \frac{dxdy}{|\hat{n} \cdot \vec{k}|}$$

$$\hat{n} \cdot \vec{k} = \frac{2}{3}$$

$$\vec{F} \cdot \hat{n} = \frac{1}{3}[2(x + y^2) - 2x + 4yz]$$

$$\iint_{R_1} \vec{F} \cdot \hat{n} ds = \iint_{R_1} \frac{1}{3}[2(x + y^2) - 2x + 4yz] \frac{dxdy}{\frac{2}{3}}$$

$$= \iint_{R_1} x + y^2 - x + 2yz dxdy$$

$$= \iint_{R_1} y(y + 2z) dxdy$$

$$\text{Put } z = \frac{6-y-2x}{2}$$

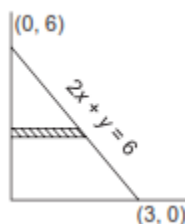
$$= \iint_{R_1} y(y+6-y-2x) dx dy$$

$$= \iint_{R_1} y(6-2x) dx dy$$

$$= 2 \iint_{R_1} y(3-x) dx dy = 2 \iint_{R_1} (3x-xy) dx dy$$

$$x \text{ varies from } 0 \text{ to } \frac{6-y}{2}$$

$$y \text{ varies from } 0 \text{ to } 6$$



$$= 2 \int_0^6 \int_0^{\left(\frac{6-y}{2}\right)} (3y-xy) dx dy$$

$$= 2 \int_0^6 \left[3yx - \frac{x^2 y}{2} \right]_0^{\frac{6-y}{2}} dy$$

$$= 2 \int_0^6 \left[3y \left(\frac{6-y}{2} \right) - \frac{1}{2} \left[\frac{(6-y)^2}{4} y \right] \right] dy$$

$$= 2 \int_0^6 \left[\frac{18y-3y^2}{2} - \frac{1}{2} \left[\frac{1}{4} [36y+y^3-12y^2] \right] \right] dy$$

$$= \int_0^6 \left[18y-3y^2 - \frac{1}{4} [36y+y^3-12y^2] \right] dy$$

$$= \left[\frac{18y^2}{2} - \frac{3y^3}{3} - \frac{1}{4} \left[\frac{36y^2}{2} + \frac{y^4}{4} - \frac{12y^3}{3} \right] \right]_0^6$$

$$= \left[9y^2 - \cancel{y^3} - \frac{9}{3}y^2 - \frac{y^4}{16} + \cancel{y^3} \right]_0^6$$

$$\begin{aligned}
&= \left[\frac{9}{2}y^2 - \frac{y^4}{16} \right]_0^6 = \left[\frac{9(36)}{2} - \frac{(36)(36)}{16} \right] \\
&= 162 - 81 \\
\iint_S \vec{F} \cdot \hat{n} ds &= 81
\end{aligned}$$

Example 3:

Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = 12x^2y\vec{i} - 3y^2\vec{j} + 2z\vec{k}$ and S is the portion of the plane $x + y + z = 1$ included in the first octant.

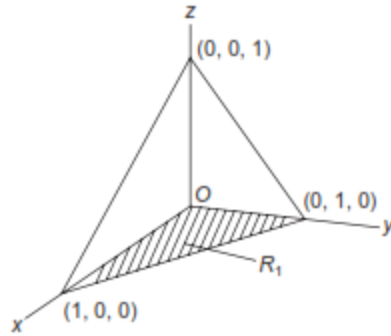
Solution:

Given, $\vec{F} = 12x^2y\vec{i} - 3yz\vec{j} + 2z\vec{k}$

$\nabla\phi = \vec{i} + \vec{j} + \vec{k}$

$\phi = x + y + z - 1$

$|\nabla\phi| = \sqrt{3}$



$$\hat{n} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$$

$$\begin{aligned}\vec{F} \cdot \hat{n} &= \frac{12x^2y}{\sqrt{3}} - \frac{3yz}{\sqrt{3}} + \frac{2z}{\sqrt{3}} \\ &= \frac{1}{\sqrt{3}}(12x^2y - 3yz + 2z)\end{aligned}$$

$\therefore$ Let R_1 be the projection of S on the xy plane

$$\therefore \quad \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{R_1} \vec{F} \cdot \hat{n} \frac{dxdy}{|\hat{n} \cdot \vec{k}|}$$

$$\begin{aligned}\iint_{R_1} \vec{F} \cdot \hat{n} \frac{dxdy}{|\hat{n} \cdot \vec{k}|} &= \iint_{R_1} \frac{1}{\sqrt{3}}(12x^2y - 3yz + 2z) \frac{dxdy}{\left|\frac{1}{\sqrt{3}}\right|} \\ &= \iint_{R_1} (12x^2y - 3yz + 2z) dxdy\end{aligned}$$

Put $z = 1 - x - y$

$$\begin{aligned}
 &= \iint_{R_1} [(12x^2y - 3y(1 - x - y) + 2(1 - x - y))] dx dy \\
 &= \iint_{R_1} [12x^2y - 3y + 3xy + 3y^2 + 2 - 2x - 2y] dx dy \\
 &= \iint_{R_1} (12x^2y - 5y + 3xy + 3y^2 - 2x + 2) dx dy
 \end{aligned}$$

x varies from 0 to $1 - y$

y varies from 0 to 1

$$\begin{aligned}
 &= \int_0^1 \int_0^{1-y} (12x^2y - 5y + 3xy + 3y^2 - 2x + 2) dx dy \\
 &= \int_0^1 \left[12y \frac{x^3}{3} - 5yx + 3y \frac{x^2}{2} + 3y^2x - \frac{2x^2}{2} + 2x \right]_0^{1-y} dy \\
 &= \int_0^1 \left[4y(1-y)^3 - 5y(1-y) + \frac{3}{2}y(1-y)^2 + 3y^2(1-y) - (1-y)^2 + 2(1-y) \right] dy \\
 &= \int_0^1 \left[4y - 12y^2 + 12y^3 + 4y^4 - 5y + 5y^2 + \frac{3}{2}y + \frac{3}{2}y^3 - 3y^2 + 3y^2 - 3y^3 - 1 - y^2 + 2y + 2 - 2y \right] dy \\
 &= \int_0^1 \left[-4y^4 + \frac{21}{2}y^3 + (-8)y^2 + \frac{1}{2}y + 1 \right] dy \\
 &= \left[-\frac{4y^5}{5} + \frac{21}{2} \frac{y^4}{4} - \frac{8y^3}{3} + \frac{1}{2} \frac{y^2}{2} + y \right]_0^1 \\
 &= \left[-\frac{4}{5} + \frac{21}{8} - \frac{8}{3} + \frac{1}{4} + 1 - 0 \right] \\
 &= -\frac{96 + 315 - 320 + 30 + 120}{120} \\
 &= \frac{465 - 416}{120}
 \end{aligned}$$

$$\iint_S \vec{F} \cdot \hat{n} ds = \frac{49}{120}$$

Example 4:

Show that $\iint_S (yz\vec{i} + zx\vec{j} + xy\vec{k}) \cdot \vec{n} \, ds = \frac{3}{8}$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant.

Solution:

Given, $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$

$$\phi = x^2 + y^2 + z^2 - 1$$

$$\begin{aligned}\nabla\phi &= \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k} \\ &= 2x\vec{i} + 2y\vec{j} + 2z\vec{k}\end{aligned}$$

$$|\nabla\phi| = \sqrt{4(x^2 + y^2 + z^2)} = \sqrt{4}$$

$$|\nabla\phi| = 2$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2(x\vec{i} + y\vec{j} + z\vec{k})}{2} = x\vec{i} + y\vec{j} + z\vec{k}$$

The projection of S on the xy plane is the circle $x^2 + y^2 = 1$ and $z = 0$

$$ds = \frac{dxdy}{|\hat{n} \cdot \vec{k}|}$$

$$\vec{F} \cdot \hat{n} = (yz\vec{i} + zx\vec{j} + xy\vec{k}) \cdot (x\vec{i} + y\vec{j} + z\vec{k})$$

$$\vec{F} \cdot \hat{n} = 3xyz$$

$$\vec{n} \cdot \vec{k} = z$$

$$\begin{aligned}\iint_S \vec{F} \cdot \hat{n} ds &= \iint_S 3xyz \frac{dxdy}{z} \\ &= \iint 3xy \, dxdy\end{aligned}$$

x varies from 0 to $\sqrt{1-y^2}$

y varies from 0 to 1

$$\begin{aligned}\iint_S \vec{F} \cdot \hat{n} ds &= \int_0^1 \int_0^{\sqrt{1-y^2}} 3xy \, dxdy \\ &= 3 \int_0^1 y \left[\frac{x^2}{2} \right]_0^{\sqrt{1-y^2}} dy \\ &= \frac{3}{2} \int_0^1 y(1-y^2) dy = \frac{3}{2} \int_0^1 (y - y^3) dy \\ &= \frac{3}{2} \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 = \frac{3}{2} \left[\frac{1}{2} - \frac{1}{4} - 0 \right] = \frac{3}{2} \left[\frac{1}{4} \right]\end{aligned}$$

$$\boxed{\iint_S \vec{F} \cdot \hat{n} ds = \frac{3}{8}}$$

Discussion Questions

1. What does a surface integral represent?
2. What is the difference between surface integral and line integrals?
3. How do you solve surface integral problems?
4. Why is surface integral zero?
5. What are the application of surface integral in vector field?